

Federico Mazzone

Cork, Ireland · mazzoni.federico1@gmail.com · [LinkedIn Profile](#) · [GitHub Portfolio](#) · [Scholar](#)

AI Engineer with experience building and deploying production LLM, RAG, and automation systems on AWS. Delivered customer-facing and internal AI products that increased engagement and reduced workflows from days to hours. Background in NLP and machine learning, with 8 peer-reviewed AI papers. EU citizen based in Ireland.

EXPERIENCE

AI Engineer

Marriott

August 2026 – Present

Cork, Ireland

- **Multi-Agent Assistant:** Worked on Q&A, folio retrieval, and human escalation for Marriott's AI assistant.

AI Engineer

Domotz

May 2024 – July 2026

Dublin, Ireland

- **Chatbot:** Engineered a FastAPI RAG-based chatbot (using Qdrant, LangChain, PostgreSQL) deployed on AWS, boosting customer engagement by 65% and serving 100+ daily users. Built multiple knowledge bases and UIs.
- **Dynamic MCP:** Developed an MCP server acting as a semantic proxy across multiple MCP servers, dynamically providing only the tools and skills relevant to each user's task via hybrid semantic search.
- **Conversation Intelligence:** Engineered an in-house Gong alternative for Sales & CS teams, featuring zero-shot LLM classification, summarization, and RAG-based chat over thousands of emails, tickets, and call transcripts.
- **Automation Systems:** Built automation systems that reduced sales, finance, and marketing workflows from days to hours, including a Slack sales bot and invoice processors handling hundreds of customers.
- **Anomaly Detection:** Combined traditional ML and LLM reasoning to detect network anomalies in time series.

PhD Research Fellow in AI

University of Pisa / National Research Council of Italy

Nov. 2022 – May 2024

Pisa, Italy

- **Research Focus:** Collaborative human-AI systems, algorithmic fairness, linguistic trust in social networks.
- Published 8 peer-reviewed articles, managed multiple concurrent projects from concept to validation, mentored university students, presented research at conferences, and organised the *Law, Society, and AI* workshops.

Freelance Data Analyst

Self-Employed

2022

Pisa, Italy

- Built recommendation systems, scraped web data for datasets, and explored bioinformatics biomarker methods.

PERSONAL PROJECTS

GAVEL: Ethical evaluation of AI model behaviour and decision-making using Game Theory (WIP).

iRAG: RAG system that learns from human decisions, reducing error while gradually taking on more autonomy.

StockAgent: Multi-agent AutoGen system for stock market analysis with a custom MCP yfinance server.

HINTT: Real-time OCR+translation tool sending screenshots to OpenAI/Ollama. Employed tkinter and mss.

Simple Palette Maker: Streamlit app to colorize black-and-white GameBoy palettes. 1k+ users.

SKILLS

AI/ML Stack: LLMs (LangChain, LangGraph, AutoGen, Bedrock), RAG, HuggingFace, SK-Learn, PyTorch, RiverML

Backend & Deployment: FastAPI, Streamlit, Docker, AWS (EC2, S3, CloudWatch, Terraform), PostgreSQL, Qdrant

Programming: Python, JavaScript, R

Platforms: Git, Jira, BitBucket, Snowflake, Zendesk, Salesforce, n8n, OpenAI, Claude, Amazon Bedrock, Azure

Languages: Italian (Native), English (C1), German (A2)

EDUCATION

University of Southern Denmark

Summer School on Deep Learning

2023

Odense, Denmark

University of Pisa

Master's Degree in Digital Humanities (Data Science and NLP) — 110/110 with honours (GPA: 4/4)

2020 – 2022

Pisa, Italy

- Relevant Courses: Data Mining, Social Network Analysis, Text Analytics, Distributed Data Analysis and Mining, Semantic Web, Laboratory of Data Science, Statistics, Applied Linguistics, Computational Linguistics.

University of Pisa

Bachelor's Degree in Philosophy — 110/110 with honours (GPA: 4/4)

2014 – 2017

Pisa, Italy